Project Report

Compost Management System

HTML, CSS, and JavaScript Web Application

Team Member 1	Ahmed Walid Aly - ID: 221257
Team Member 2	Eslam Samir - ID: 204581
Team Member 3	Omar Walid - ID: 200927
Course	CSE5531 | Web Design Concepts
Instructor	Dr. Ahmed Ayoub
Department	Computer Systems Engineering, MSA University

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1. Executive Summary

The Compost Management System is a simple web application designed for farmers who need an organized way to track and manage composting activities for soil enrichment. The system allows the farmer to add compost batches, record the type of organic material, monitor temperature, moisture, pH, composting stage, and the next turning date. It also gives visual dashboard statistics and automatic alerts when a batch needs attention.

The project is implemented using HTML, CSS, and JavaScript only. JavaScript is used for form handling, validation, dynamic interface updates, health analysis, search and filtering, CSV export, and localStorage saving. The result is a lightweight system that can run directly in the browser without a server.

2. Introduction

Composting is an important farming activity because it converts organic waste into natural fertilizer. Good compost improves soil structure, increases nutrients, and reduces the need for chemical fertilizers. However, composting needs regular monitoring. If moisture is too low, decomposition slows down. If temperature is too high, useful organisms may be harmed. If the pile is not turned on time, the compost quality can drop.

The proposed system solves this problem by providing a clean digital interface for farmers. Instead of using paper notes, the farmer can save each compost batch in the browser and check its condition at any time. The system is suitable for small and medium farms because it is easy to use and does not require advanced technical knowledge.

3. Project Objectives

- Create a basic compost management web system using HTML, CSS, and JavaScript.
- Allow farmers to add, edit, delete, search, and filter compost batches.
- Track important compost values such as temperature, moisture, pH, stage, and turning date.
- Display dashboard statistics for total batches, active batches, ready batches, and batches needing attention.
- Generate simple advice and alerts based on compost health conditions.
- Store data using browser localStorage so the records remain after refreshing the page.
- Provide CSV export so the farmer can keep a copy of the compost records.

4. Project Scope

The scope of the project is a front-end web application. It does not use a database server or back-end framework. The system focuses on the main tasks a farmer needs in a basic compost management workflow: batch creation, batch monitoring, health checking, reminders, and record exporting.

Included Features

- Responsive dashboard page.
- Compost batch form with required fields.
- Automatic compost health analysis.
- Search and stage filter.
- Edit and delete actions.
- CSV export.
- Browser-based data persistence using localStorage.

Not Included in Current Version

- Real IoT sensor connection.
- Online user login system.

- Cloud database synchronization.
- Mobile application version.

5. Technologies Used

Technology	Purpose
HTML5	Builds the structure of the web page and input form.
CSS3	Creates the layout, colors, cards, responsive design, and visual styling.
JavaScript	Handles logic, localStorage, validation, alerts, dashboard statistics, editing, deleting, filtering, and CSV export.
Browser localStorage	Stores compost records locally inside the user browser.
CSV Export	Allows the farmer to download saved compost data as a spreadsheet-compatible file.

6. System Design

The system is designed as a single-page web application. The farmer interacts with the interface, enters compost batch data, and JavaScript processes the data. The data is stored locally in the browser. The dashboard, alerts, and batch cards are updated dynamically without reloading the page.

6.1 System Architecture

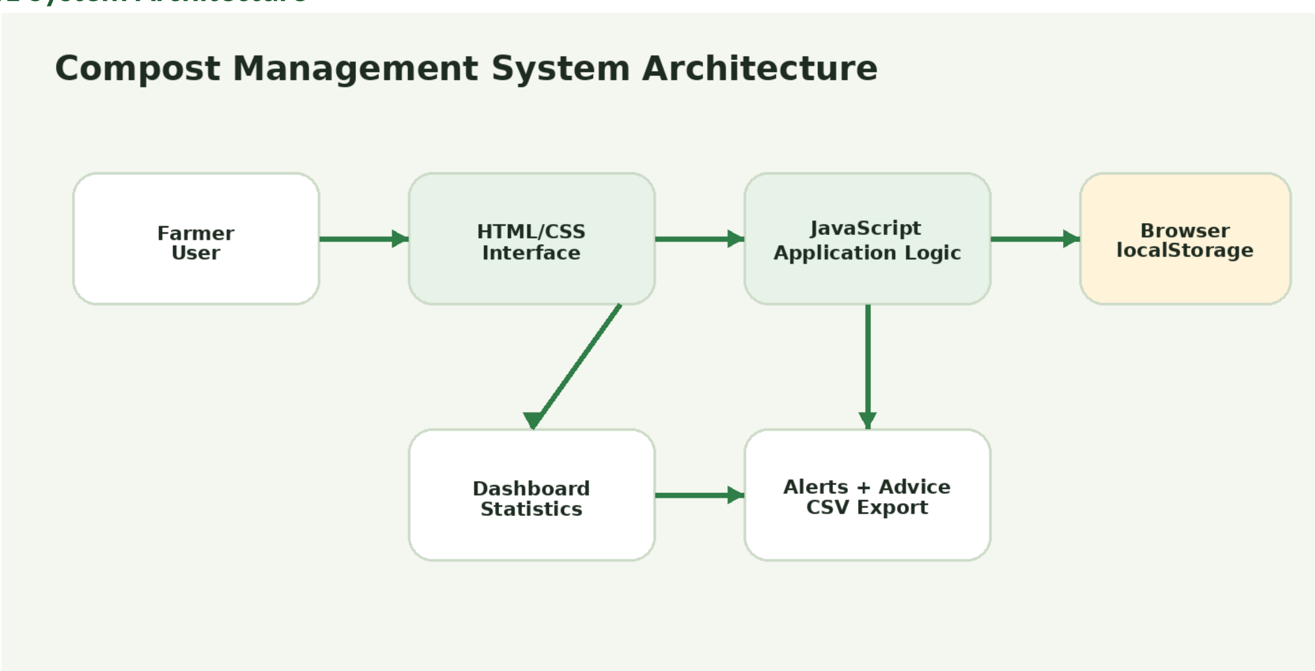

Figure 1. Compost Management System architecture.

6.2 Main Workflow

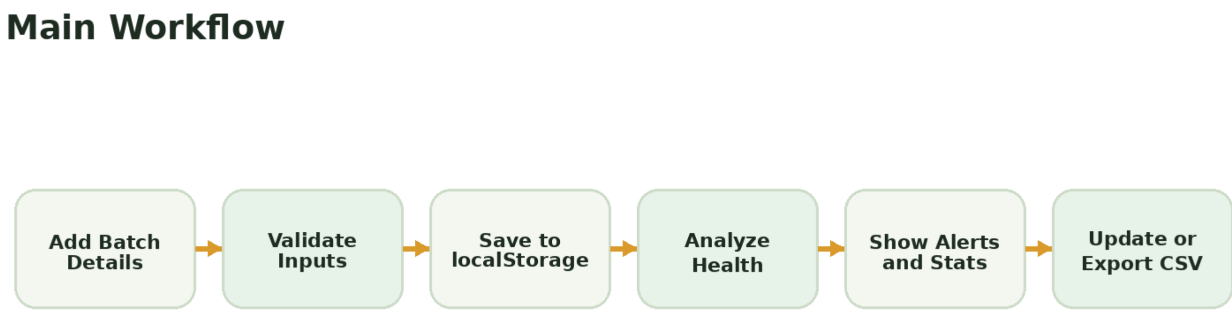

Figure 2. Main workflow of the Compost Management System.

6.3 Data Model

Field	Description
Batch Name	Name used to identify the compost batch.
Organic Material	Main material used, such as vegetable waste, crop residue, manure, or leaves.
Start Date	Date when the compost batch started.
Stage	Current compost stage: Fresh, Active, Curing, or Ready.
Temperature	Current temperature in Celsius.
Moisture	Current moisture percentage.
pH	Acidity or alkalinity value of the compost.
Next Turning Date	Date when the farmer should turn the compost pile.
Notes	Short farmer observations about smell, color, texture, or actions.

6.4 Health Rules

Compost Factor	Good Range	System Action
Temperature	45°C to 65°C	Warns if the temperature is too low or too high.
Moisture	40% to 60%	Warns if the pile is too dry or too wet.
pH	6.0 to 8.0	Warns if the pH is outside the suitable range.

Turning Date	Not overdue	Warns when the next turning date is due.

7. User Interface Design

The improved user interface was designed to be clean, organized, and clearly connected to the compost management idea. It uses a green and earthy farming theme, soft rounded cards, clear spacing, dashboard panels, a compost health ring, priority alerts, and readable batch cards. The layout is responsive, so it works on laptop, tablet, and mobile screens.

- The top section contains a professional hero area with the project purpose, current date, compost health ring, and quick overview values.
- The dashboard cards summarize total batches, active batches, ready batches, batches needing attention, turning due count, and overall compost health.
- The form panel is kept on the side of the dashboard to let the farmer add or update compost records quickly with clear input fields.
- The compost batch cards show each record using status badges, health chips, maturity progress bars, temperature, moisture, pH, and next turning date.
- The priority alerts section highlights urgent compost problems and gives simple action advice for the farmer.
- The farmer guide and lifecycle sections explain the healthy compost ranges and the main compost stages: Fresh, Active, Curing, and Ready.

8. Implementation

8.1 File Structure

File	Role
index.html	Contains the page structure, top bar, hero section, dashboard cards, form, batch list, alerts, guide section, and lifecycle section.
style.css	Contains the improved farming theme, responsive layout, cards, health ring, progress bars, alerts, and visual styling.
app.js	Contains all JavaScript logic for data handling, storage, dashboard updates, health calculation, alerts, search, filtering, editing, deleting, and CSV export.

8.2 JavaScript Functions

- loadBatches() loads saved compost data or creates sample records.
- saveBatches() stores the compost data in localStorage.
- isHealthy() checks if the compost values are inside the correct ranges.
- needsAttention() detects batches that need farmer action.
- getAdvice() generates simple advice for compost problems.
- renderBatches() displays filtered compost batch cards with status badges, progress bars, and health chips.

- updateStats() updates dashboard values, compost health percentage, health label, and turning due count.
- exportCsv() downloads compost records as a CSV file for backup or spreadsheet use.

8.3 Validation

The system uses required input fields and numeric limits for temperature, moisture, and pH. This reduces wrong input and keeps the compost data more reliable.

9. Results and Discussion

The project successfully provides a functional compost management interface. The farmer can create new records, save them, edit them, delete them, search through them, filter by stage, and export the records. The dashboard updates immediately after every action.

9.1 Testing Summary

Test Case	Expected Result	Status
Add a new compost batch	The batch appears in the list and is saved.	Passed
Edit an existing batch	The form loads the selected batch and updates it.	Passed
Delete a batch	The selected batch is removed from the list.	Passed
Search by name or material	Only matching batches are displayed.	Passed
Filter by stage	Only batches with the selected stage are displayed.	Passed
Enter unhealthy temperature/moisture/pH	The system displays an alert and advice.	Passed
Refresh the page	Saved batches remain available using localStorage.	Passed
Export CSV	A CSV file is downloaded with compost records.	Passed

9.2 Discussion

The system is useful because it turns compost monitoring into a simple digital workflow. The automatic alerts make the project stronger than a normal record form because the system checks the data and gives farmer-friendly advice. The use of localStorage is also suitable for a basic coursework project because it avoids back-end complexity while still saving user records.

10. Conclusion

The Compost Management System meets the main requirement of creating a basic web-based system for farmers to track and manage composting activities. It supports compost batch management, health monitoring, reminders, alerts, and data export. The project is simple, practical, and directly connected to soil enrichment and smart farming.

11. Future Work

- Add real IoT sensors to read temperature and moisture automatically.
- Add user accounts and cloud database storage.
- Add charts to show compost temperature and moisture history over time.
- Add Arabic language support for local farmers.
- Add notification reminders for turning dates.
- Add image upload for compost pile observations.
- Create a mobile app version for field use.

12. Evaluation Rubric Alignment

Rubric Area	How the Project Supports It
Functionality	The project performs the required compost tracking tasks and includes add, edit, delete, search, filter, alerts, localStorage, and CSV export.
User Interface	The improved GUI uses a clean farming theme, responsive dashboard layout, compost health ring, stat cards, priority alerts, batch cards, progress bars, status badges, and guide/lifecycle sections.
Code Quality	The code is separated into HTML, CSS, and JavaScript files with clear variable names, organized functions, and no unnecessary complexity.
Innovation and Problem Solving	The system adds automatic health checking, farmer advice, and CSV export instead of only storing records.

13. Challenges and Solutions

Challenge	Solution
Keeping the interface simple for farmers	The system uses a clear dashboard, themed cards, simple labels, visual status indicators, and direct buttons instead of complex menus.
Saving data without a database	The project uses browser localStorage to keep records after page refresh.
Helping farmers understand compost problems	JavaScript checks compost values and generates simple advice.
Making the page work on different screens	CSS media queries and flexible grid layouts make the improved GUI responsive on desktop, tablet, and mobile screens.
Avoiding unnecessary complexity	The project stays front-end only and focuses on the required coursework scope.

14. Recommendations

- Use the system daily or weekly to update compost temperature, moisture, and pH values.
- Check the alert section before applying compost to soil.
- Keep turning dates updated to maintain good oxygen levels inside the pile.
- Export CSV records regularly as a backup.
- Use the ready stage only when the compost has an earthy smell, stable temperature, and mature texture.

15. How to Run the Project

1. Open the project folder named compost_management_system.
2. Open index.html using any modern web browser.
3. Add or update compost batches using the form.
4. Use the dashboard, alerts, search, filter, and CSV export features.